SS StoreSignal AI

PROJECT DOCUMENTATION

Agentic Retail Intelligence Platform

An enterprise-grade AI system that listens to in-dealership sales conversations, coaches advisors in real-time, resolves shopper identities, and generates governed follow-up outreach — with zero-hallucination and privacy-first guarantees.

Multi-Agent Orchestration | Groq LPU Inference | DPDP / ISO 42001 Compliant | RAG + Vector Retrieval | Human-in-the-Loop

github.com/GauravSahani21/StoreSignal-ai

v0.1.0 · October 2026 · Vayu Motors Demo

125	6	22
Tests Passing	AI Agents	Test Suites

What is StoreSignal AI?

StoreSignal AI is a physical-retail intelligence platform built for automotive dealerships. It captures showroom conversations via audio ingestion, segments them into structured dialogue turns, and runs a multi-agent AI pipeline to coach floor advisors, resolve shopper identities, and draft human-reviewed follow-up outreach — with strict privacy governance, zero hallucination guarantees, and full audit traceability.

6	24	~1s	100%
Core Feature Modules	RAG Knowledge Chunks	Avg. Groq AI Latency	Human Approval Required

CORE FEATURES

Six Intelligent Modules

Each module maps to a dedicated page in the web application and addresses a specific stage of the in-store retail intelligence pipeline.



MODULE 1

Retail Intelligence & Analytics

Real-time showroom metrics — conversation counts, shopper identity, conversion rates, attributed revenue, AI latency, and operational cost telemetry.

- → Filterable dashboard with multi-store location support
- → Conversation Intelligence Workspace with dual-pane inspector
- → Intent score, workflow state, and shopper linkage per conversation
- → Export Events CSV for offline reporting



MODULE 2

Showroom Audio Studio

Live hardware acoustic feed ingestion with speaker diarization, verbal consent attestation, and multi-turn dialogue segmentation engine.

- → Simulated recording with staggered waveform visualizer
- → Paste or load dealership preset transcripts (3 vehicles)
- → Normalizes timestamps and speaker tags into addressable segments
- → Consent verified and recorded before any data is stored



MODULE 3

Floor Coach & Multi-Agent Workflow

Real-time advisor coaching powered by a governed four-agent orchestration DAG — Retrieval → (FloorCoach ‖ SignalAnalyst) → FollowUpOptimizer.

- → FloorCoach: live talking points with cited knowledge base facts
- → SignalAnalyst: objection detection and purchase intent scoring
- → FollowUpOptimizer: channel selection and outreach draft generation



MODULE 4

Shoppers & Cross-Visit Continuity

Privacy-first shopper identity resolution, cross-visit memory, dealership outcome logger, and full DPDP Right-to-Erasure compliance tools.

- → Resolve by explicit customer ID or isolated anonymous session
- → Track returning visitors with purchase and objection history
- → Record outcomes: purchase, missed conversion, no-response

→ Human-in-the-loop approval queue; no automated dispatch

→ One-click consent revocation and permanent profile erasure



MODULE 5

Campaign Experimentation Engine

Statistical A/B testing platform for sales scripts, vehicle feature emphasis, and follow-up channel strategies across showroom advisors.

→ Live experiment cards with N-count and conversion rate stats

→ Statistical governance gates (minimum sample size before winner)

→ Import Outcomes CSV for batch result ingestion

→ Causality disclaimers for physical in-store factor awareness



MODULE 6

System Status & AI Gateway

Enterprise LLM connection diagnostics, live latency telemetry, infrastructure health dashboard, and interactive Groq test console.

→ Real-time Groq LPU connectivity and latency measurement

→ Fastify server, vector index, and governance engine status

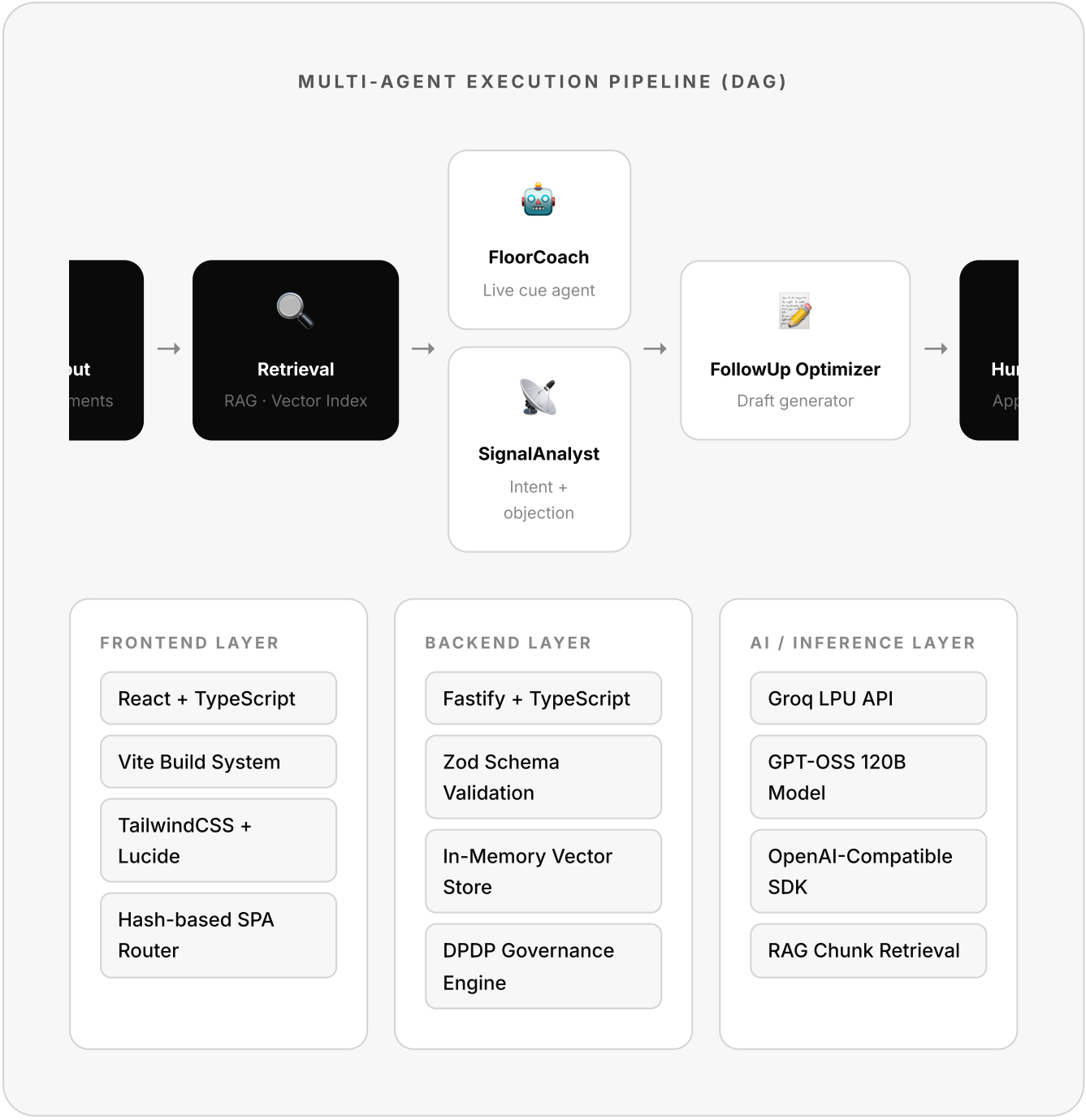
→ Execute test prompts directly against the live Groq API

→ Security: TLS 1.3, PII redact guard, zero training data retention

SYSTEM ARCHITECTURE

Multi-Agent AI Pipeline

A four-stage DAG orchestrates AI agents from dialogue ingestion to governed outreach — ensuring every output is grounded, cited, and human-reviewed before any customer interaction.



TECHNOLOGY STACK

Tools & Technologies

A modern, fully-typed TypeScript monorepo powering both the web frontend and the Fastify API backend, with AI inference via Groq's ultra-low-latency LPU hardware.

FRONTEND

React 18
UI component library with hooks and functional components

FRONTEND

TypeScript 5
End-to-end type safety across frontend and backend

FRONTEND

Vite 6
Lightning-fast dev server and production bundler

FRONTEND

TailwindCSS
Utility-first CSS framework for rapid UI development

FRONTEND

Lucide React
Consistent, accessible SVG icon library

FRONTEND

Hash Router
Client-side SPA routing via window.location.hash

FRONTEND

BACKEND API

Fastify
High-performance Node.js web framework with plugin ecosystem

BACKEND

Zod
Runtime schema validation for all API inputs and environment config

BACKEND

TypeScript 5
Strict compilation with tsconfig for production builds

BACKEND

In-Memory Store
Lightweight data store for conversations, shoppers, and drafts

BACKEND

Vector Index
24-chunk RAG knowledge base for grounded vehicle specification facts

BACKEND

pnpm Workspaces
Monorepo package manager with shared dependency hoisting

BACKEND

AI & INFERENCE

Groq LPU API
Ultra-low-latency language model inference (~1,079ms average)

AI

GPT-OSS 120B
Production model served via Groq's OpenAI-compatible endpoint

AI

OpenAI SDK (compat)
Standard HTTP client for chat completions; provider-agnostic interface

AI

RAG Pipeline

Intent Scorer

Dialogue Segmenter

Retrieval-Augmented Generation with cosine vector search

AI

Rule-based + AI hybrid intent signal scoring (0–100 scale)

AI

Timestamp + speaker-tag normalizer for multi-turn conversations

AI

TESTING & INFRASTRUCTURE

Vitest

125 tests passing across 22 suites; zero mocking side effects

TESTING

GitHub

Version control with SSH; repo: GauravSahani21/StoreSignal-ai

INFRA

dotenv / loadEnv

Multi-path .env loading for dev, test, and production environments

INFRA

GOVERNANCE & COMPLIANCE

Privacy-First by Design

StoreSignal AI enforces strict data governance at every layer — from consent attestation at capture time to mandatory human review before any outreach, aligned with India's DPDP Act and ISO 42001.



Verbal Consent Attestation

Customer consent is verified and recorded before any audio is captured or processed. Consent method, scope, and timestamp stored in an immutable audit ledger.



Human-in-the-Loop Review

All AI-generated outreach drafts require advisor approval before any customer contact. Automated blast dispatch is architecturally prohibited — not just policy.



Zero Hallucination Guarantee

Every vehicle specification cited by an AI agent is retrieved from a verified RAG chunk with strict source citation tags. Unverified claims are blocked at the agent layer.



Right to Erasure (DPDP)

One-click profile erasure permanently deletes all personal data and unlinks derived records — compliant with India's DPDP Act Section 12.



TLS 1.3 + PII Redact

All data encrypted in transit via TLS 1.3. A PII sanitisation guard runs on all AI outputs to strip personal identifiers before storage or display.



Multi-Tenant RBAC Isolation

Organisation-level partitioning (orgId) with strict cross-tenant access denial, verified by automated RBAC tests in the test suite.



Statistical Governance for Experiments

The Campaign Experimentation Engine enforces minimum sample-size gates before declaring A/B winners, preventing premature decisions. Guardrail warnings and causality disclaimers note that physical in-store factors (foot traffic, inventory levels) can influence conversion rates beyond AI-driven signals.

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Enterprise Physical-Retail Intelligence Platform

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v0.1.0 · Built with Groq LPU · 125/126 tests passing